



Normative Recommenders and News Consumption

Impact of Diversity-Optimized Recommendations on News Reading Behavior and Political Positions

Lucien Heitz, University of Zurich

Juliane Lischka, University of Hamburg

Bibek Paudel, Stanford University

Suzanne Tolmeijer, University of Zurich

Alena Birrer, University Zurich

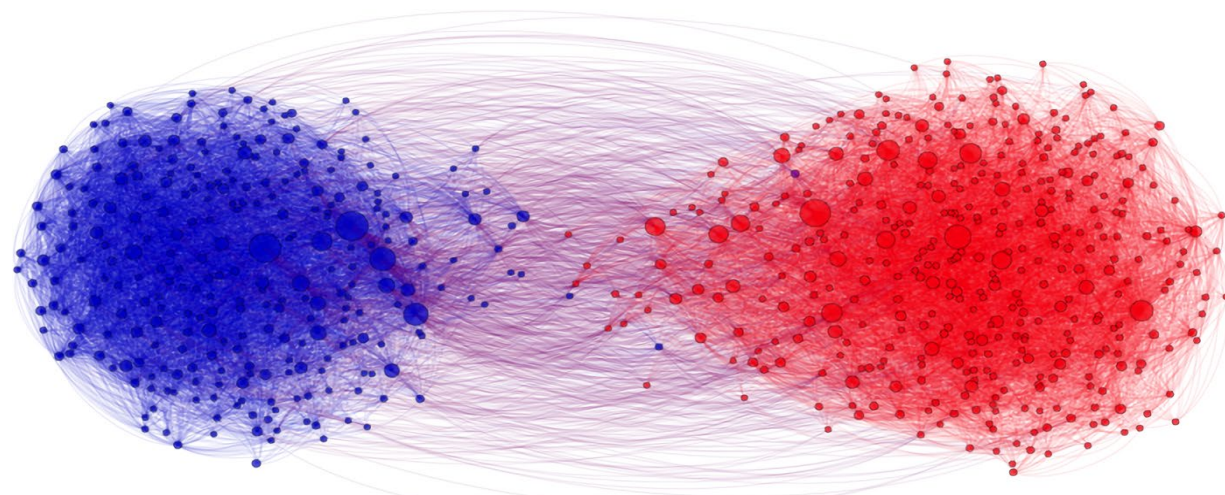
Abraham Bernstein, University of Zurich



Introduction and Motivation

Role of Diversity

- Viewpoint diversity in media is a crucial part of democracy
- Broad use of personalized recommenders on online platforms
- Danger of polarizing the readership



Visualizing Polarization in Political Blogs

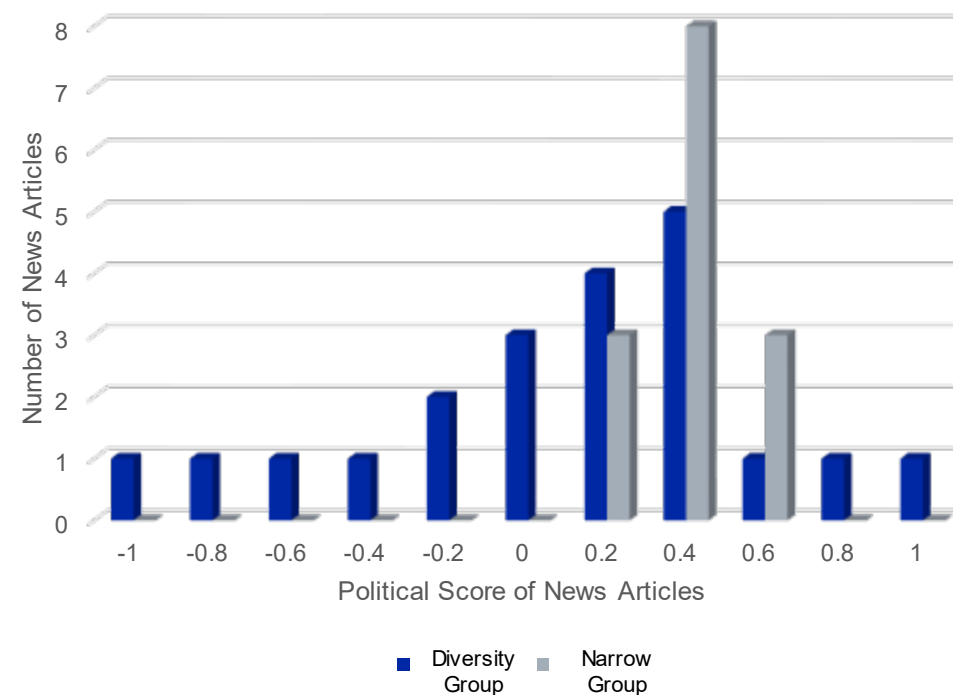
<http://allthingsgraphed.com/2014/10/09/visualizing-political-polarization/>



Operationalizing Diversity

- Diversity is always **normative** and there exist **multiple** definitions
- Political diversity depends on the interpretation of democracy theory and its goals
- Combination of participative and deliberative approaches focusing on the center
- Diversity defined as **discretized article distribution** in an n-dimensional space

Sample distribution of news articles for a user with a score of 0.4 (left to right spectrum).



Bund ruft wegen Hitzewelle Gefahrenstufe 4 auch für das Tessin aus – erste Juni-Rekorde purzeln

Heute, 11:42 Uhr – 6 Min. Lesezeit



Kanadier haben die Schönste: Neues 200er-Nötli war chancenlos

Jahr für Jahr werden die schönsten...

Heute, 18:30 Uhr – 1 Min. Lesezeit



An diesen 10 kuriosen Automaten können Sie auch am 1. Mai einkaufen: CBD, Pouletschenkel oder...

Heute, 17:04 Uhr – 4 Min. Lesezeit



Kanadier haben die Schönste: Neues 200er-Nötli war chancenlos

Jahr für Jahr werden die schönsten Banknoten
der Welt prämiert. Für einmal musste sich die
Schweiz geschlagen geben.

Heute, 18:30 Uhr



Diesmal hats nicht gereicht: Die kanadische 10-
Dollar-Note mit der Bürgerrechtspionierin Viola
Desmond wurde zum schönsten Geldschein der
Welt gekürt. Das hat die Internationale



Lesezeichen (3)

Archiv

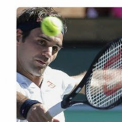
Die letzte Zuflucht für Simbabwe wilde Tiere



Montag, 18.03.2019 - 7 Min. Lesezeit



Indian Wells: Roger Federer gewinnt den ersten Satz gegen Hubert Hurkacz...



Freitag, 15.03.2019 - 1 Min. Lesezeit



Wo China von Träumen lebt – ein Foto-Tableau von Ronghui Chen

China ist der Wirtschaftsgigant, vor dem...



Montag, 11.03.2019 - 5 Min. Lesezeit



Wie oft nutzen Sie News-Apps auf Ihrem Smartphone?

Bitte wählen Sie eine Antwort

Jeden Tag.

Mehrmals pro Woche.

Ich nutze keine News-Apps.

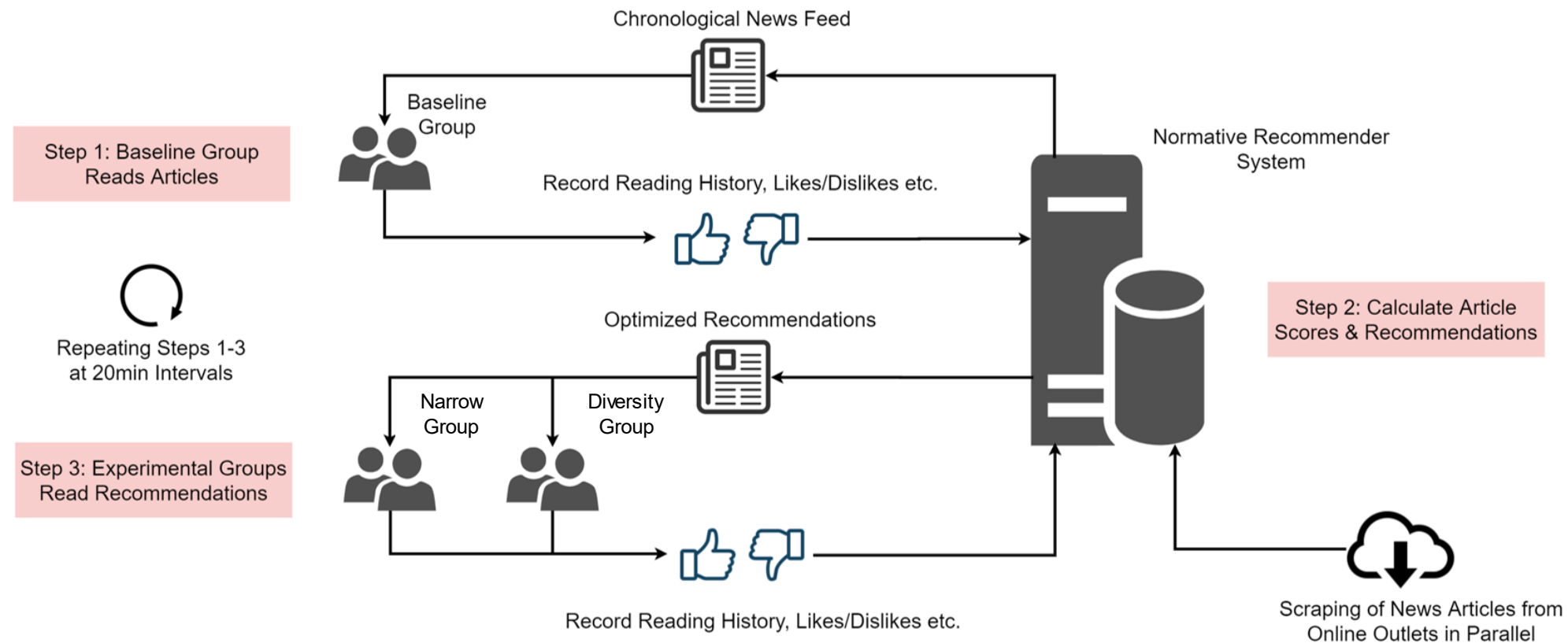
Zurück

Abschliessen



UZH

DDIS
NEWS





Impact on User Utility (RQ1)

Evaluation on the Basis the Exit Surveys

- Better news overview and lower use of external news sources (diversity vs. narrow, right-leaning users)
- Higher appreciation of journalism (diversity vs. narrow) and content diversity (heavy vs. medium users)

Evaluation on the Basis of Interaction Metrics

- Diversity shows tendency to positively impact user engagement with a delicate trade-off between likes and dislikes
- Diversity matches and possibly outperforms current strategies

	Base.	Nar.	Div.
Articles	271	211	313
Time	6.8h	5.3h	7.0h
Likes	247	171	292
Dislikes	45	61	50
Ratio	5.5	2.8	5.8

Impact of different news recommendation strategies on user engagement rates across groups (averages, n.s.).



Impact on Political Position (RQ2)

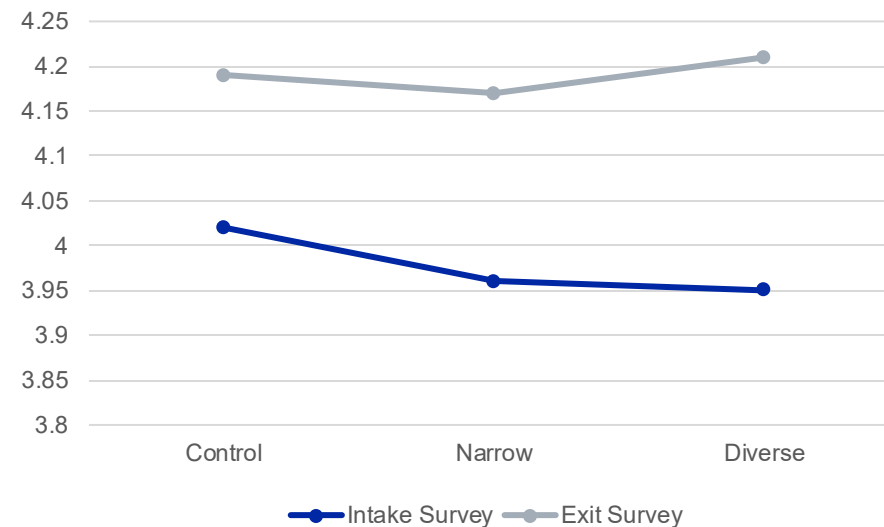
Evaluation on the Basis of Intake vs. Exit Surveys

- **Significant change** of the political score in the diversity group
- Diversity reduces the number of people with extreme scores
- Changes follow normative distribution of the algorithm

Evaluation on the Basis of Interaction Metrics

- Normal distribution of **read** articles for baseline group and a flat distribution of articles for the diversity group
- Distribution of read articles follows again the algorithm

Change in political position on a scale from 1 (left), 4 (center) and 7 (right).





University of
Zurich^{UZH}

Department of Informatics – Dynamic and Distributed Information Systems Group



Become a part of this project.
Download the app now!

For more information:
ddis-news-app.ifi.uzh.ch/info
ddis-news-app@ifi.uzh.ch

Thank you very much for your attention!

Contact: Abraham Bernstein (bernstein@ifi.uzh.ch)
Lucien Heitz (heitz@ifi.uzh.ch)